

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.  
seq1 <- seq(1, 55, 5)  
seq1
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
# Sequence 1-55, intervals of 5 named 'seq1'
```

```
#2.  
mean(seq1)
```

```
## [1] 26
```

```
# I computed the mean of the 'seq1' I created  
median(seq1)
```

```
## [1] 26
```

```
# I computed the median of the 'seq1' I created
```

```
#3.
```

```
mean(seq1) > median(seq1)
```

```
## [1] FALSE
```

```
# Conditional statement mean > median?
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
CharacterType_Names <- c("Lucy", "Joe", "Bob", "Mary")
```

```
NumericType_TestScores <- c(75, 89, 95, 47)
```

```
LogicalType_Scholarship <-c(FALSE, TRUE, TRUE, FALSE)
```

```
SchoolRoster <- data.frame(CharacterType_Names,NumericType_TestScores,LogicalType_Scholarship)
```

```
names(SchoolRoster) <- c("Name","Test Score","Receives Scholarship")
```

```
SchoolRoster
```

```
##   Name Test Score Receives Scholarship
## 1 Lucy         75             FALSE
## 2  Joe         89              TRUE
## 3  Bob         95              TRUE
## 4 Mary         47             FALSE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: This data frame is different from a matrix because it contains different data types (character, numeric, logical) for the different columns. A matrix would have to have uniform data types for the columns.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word "Pass"; otherwise print the word "Fail".
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```

#10. Create a function using if...else
function1 <- function(x) {
  if(x > 50) {
    print("Pass")
  }
  else {
    print("Fail")
  }
}

#11. Create a function using ifelse()
function2 <- function(x){
  ifelse(x>50,"Pass","Fail")
}

#12a. Run the first function with the value 52.5
answer12a <- function1(52.5)

```

```
## [1] "Pass"
```

```
answer12a
```

```
## [1] "Pass"
```

```

#12b. Run the second function with the value 52.5
answer12b <- function2(52.5)
answer12b

```

```
## [1] "Pass"
```

```

#13a. Run the first function with the vector of test scores
#answer13a <- function1(NumericType_TestScores)
#13b. Run the second function with the vector of test scores
answer13b <- function2(NumericType_TestScores)
answer13b

```

```
## [1] "Pass" "Pass" "Pass" "Fail"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The option ‘`ifelse`’ worked because it is vectorized. The ‘`if...else`’ option is not vectorized and works on single conditions rather than a vector input, which is why it did not work.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)